

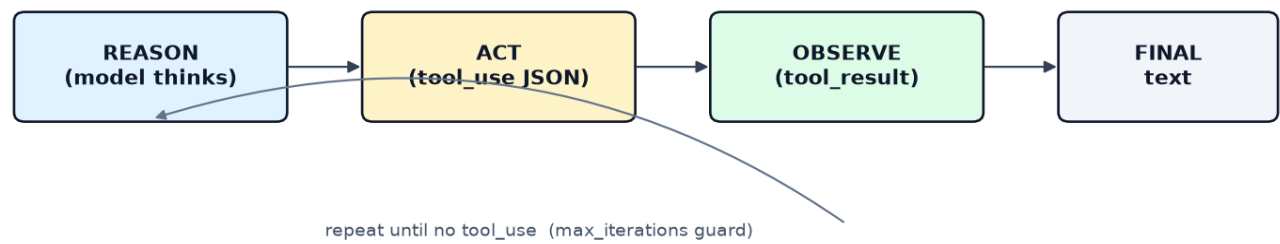
Week 2 Day 1 — Agent Foundations (Raw Python Tool Calling & ReAct)

Mental Model: An agent is an LLM in an execution loop: it reasons, selects tools, observes results, and decides what to do next. A chatbot answers in a single shot without tools; a workflow executes a hardcoded sequence of steps. Agentic systems require autonomy, tool use, multi-step planning, and self-correction. *Overkill:* An agent is unnecessary for deterministic lookups, static FAQs, or known DAGs where a basic script or prompt is faster, cheaper, and predictable.

ReAct loop

Reason (optional text) → **Act** (`tool_use JSON`) → **Observe** (`tool_result`) → repeat until the model emits a final text turn. The Python `while` loop in `agent.py` is that cycle, with `max_iterations=8` as a hard stop.

ReAct loop — Reason → Act → Observe → repeat



Tool schemas

Descriptions are the model's API docs. Vague names cause hallucinated arguments; precise descriptions tell the model *when* to call, *what* type to send, and *what* the tool cannot do.

Tool	Inputs	Role
calculator	expression: string	Safe AST arithmetic (+ − × ÷ **). Not Python eval.
get_weather	city: string	Stub Celsius lookup (Karachi, London, Tokyo, NY, Oslo).
read_sandbox_file	filename: string	Read sandbox/ only; blocks path traversal.
broken_lookup	query: string	Always raises — used to study error handling.

Worked example: “weather in Karachi and London — which is warmer?” → two `get_weather` calls → optional `calculator` on 33-12 → final answer (Karachi warmer by 21 C).

Failure modes observed & mitigations

Failure	Mitigation
Infinite Reason–Act loop	<code>max_iterations</code> hard stop; log every step.
Hallucinated tool name	Dispatch table; return <code>ERROR</code> for unknown names; <code>is_error=True</code> .
Wrong / unsafe tool arguments	JSON schema + required fields; calculator uses AST, not eval.
Silent tool exceptions	Catch, return <code>ERROR:..., log [observe]</code> ; model must not invent numbers.
Ambiguous user request	Ask a clarifying question instead of guessing a city/file.
Missing tool (e.g. stock price)	Refuse to fabricate; tell the user the tool does not exist.

Why LangChain / LangGraph / CrewAI exist

The hand-rolled loop already works, but production agents need retries, tracing, structured output parsing, graph control flow, human-in-the-loop gates, memory stores, and multi-agent routing. Frameworks are conveniences around this same ReAct core — not a different kind of intelligence. After Day 1, those libraries should read as wrappers over messages, tools, and a stop condition.

Reproduce: `pip install -r requirements.txt · set GEMINI_API_KEY in .env · python agent.py`